

In-plant management for pollution reduction

In-plant pollution management comprises a range of practices designed to minimize the environmental impact of industrial operations. These practices include maintaining proper housekeeping, segregating waste streams, reducing water consumption, optimizing chemical use, promoting the recycling and reuse of materials, and monitoring the overall water footprint of the facility.

Key elements of in-plant pollution reduction

Housekeeping

Maintaining a clean and organized workspace helps identify potential spills, leaks, and areas for improvement in waste management and resource efficiency.

Proper storage of chemicals

Store chemicals safely to prevent leakage and accidental discharge.

Process control

Maintain optimum temperature, pH, time, chemical dosage, and other operating conditions.

Raw material optimization

Use high-quality raw materials and minimize material losses during processing.

Segregation of waste streams

Separating different types of waste (hazardous, recyclable, organic) allows for proper disposal and recycling, reducing overall waste volume and potential contamination.

Reduction of water use

Implementing measures like installing water-efficient fixtures, fixing leaks, and using recirculating cooling systems can significantly decrease water consumption and associated wastewater discharge.

Chemical use reduction

Minimizing the use of hazardous chemicals through process optimization, substitution with less harmful alternatives, and proper storage can reduce chemical spills and pollution.

Equipment maintenance

Regularly inspect and maintain machinery to prevent leaks and inefficient operation.

Energy management

Reduce unnecessary energy consumption and emissions through efficient equipment and processes.

Recycling and reuse of chemicals and water

Implementing systems to collect, treat, and reuse wastewater and chemicals internally can significantly reduce discharge and resource consumption.

Worker training

Train workers in pollution-prevention practices, chemical handling, and waste management.

Water footprint

Monitoring and managing the water footprint helps identify areas for improvement and ensure responsible water usage throughout the production process.

Benefits of in-plant pollution reduction:

Reduced environmental impact

Minimizing pollution discharge protects natural resources, improves air and water quality, and contributes to sustainable development.

Cost savings

Implementing efficient practices can lead to reduced energy consumption, water usage, and waste disposal costs.

Conservation of resources

Promotes efficient use of water, chemicals, energy, and raw materials.

Improved compliance

Adhering to environmental regulations and standards avoids penalties and legal issues.

Recovery of valuable materials

Useful chemicals and materials can be recovered and reused.

Improved workplace safety

Proper chemical handling, storage, and housekeeping reduce workplace hazards.

Enhanced brand reputation

Companies that prioritize sustainability can attract customers and improve their brand image.

Housekeeping

Housekeeping in an industrial context is systematic care and management of equipment, materials, and workplace practices to ensure efficiency, safety, and pollution prevention. It goes beyond just “cleaning” – it is about discipline, preventive actions, and process integrity to reduce waste generation and environmental impact. Housekeeping is defined as the practice of maintaining operational discipline and carrying out regular maintenance activities to avoid unnecessary waste and pollution. It covers not only the cleanliness of the workplace but also the proper management of raw materials, chemicals, equipment, and processes. A well-implemented housekeeping program ensures that resources are used efficiently, environmental risks are minimized, and the plant operates smoothly without disruptions.

The primary objectives of housekeeping are to prevent losses of raw materials, water, chemicals, and energy through leaks, spills, or poor practices, and to minimize the risk of pollution. It also focuses on improving worker safety and occupational health by ensuring a clean and hazard-free workplace. Additionally, good housekeeping enhances equipment reliability, reduces downtime, and ensures compliance with regulatory requirements, which ultimately improves productivity and reduces operational costs.

Key elements of good housekeeping

i) Material Handling and Storage

Improper storage and handling of raw materials and chemicals are a major cause of workplace pollution. Materials should be stored in dedicated, well-ventilated, and clearly marked storage areas with spill-proof flooring and secondary containment (bunds, trays, double-walled tanks) to prevent leaks from contaminating soil or water. Hazardous substances must be labeled with globally harmonized system (GHS) symbols and Material Safety Data Sheets (MSDS) should be accessible for worker guidance. A First-In-First-Out (FIFO) or First-Expired-First-Out (FEFO) system minimizes waste from expired chemicals.

ii) Leak and Spill Management

Leaks and spills are common in plants dealing with liquids, gases, or fine powders, and they are a hidden source of both economic loss and environmental damage. Installing drip trays and splash guards under pumps, tanks, and valves ensures that even minor leaks are contained. For pipelines carrying hazardous chemicals, automatic shut-off valves and double seals can prevent large accidental releases. Industries also keep spill kits with absorbent pads, sand, sawdust, or activated carbon at strategic locations for immediate response. Moreover, workers are trained to act quickly, as even small leaks can accumulate into significant losses over time.

iii) Equipment Maintenance

Housekeeping strongly overlaps with equipment management because poorly maintained equipment is one of the biggest sources of pollution and waste. Preventive maintenance schedules—covering lubrication, calibration, cleaning, and inspection—keep equipment

functioning efficiently and reduce breakdowns that could lead to leaks or emissions. Predictive techniques like vibration analysis, infrared thermography, and ultrasonic testing allow plants to detect problems before they cause environmental or safety hazards. For example, detecting a small crack in a pipe early prevents costly chemical leaks later.

iv) Workplace Cleanliness

Maintaining workplace cleanliness is another cornerstone of good housekeeping. Clean floors, uncluttered spaces, and organized storage areas reduce the chances of accidents and contamination. For cleaning, industries are moving away from high-water-use methods and shifting to high-pressure, low-volume systems, mechanical sweepers, and vacuum-based systems to minimize both water use and wastewater generation. Separating areas for hazardous and non-hazardous waste helps prevent cross-contamination, while color-coded bins and floor markings ensure that workers follow waste management rules correctly. A tidy and organized workplace also improves morale and efficiency, as workers can perform tasks without obstruction or confusion.

v) Worker Training and Awareness

Workers are the most critical link in the housekeeping chain. Even the best equipment and systems fail if employees are not trained to follow procedures. Regular training sessions on safe chemical handling, proper storage, emergency response, and waste minimization instill a culture of safety and pollution prevention. Simple actions, such as reporting a small leak immediately or using the right protective gear, can prevent larger environmental hazards. Visual aids such as posters, signage, and safety charts reinforce awareness. In advanced plants, incentive programs reward workers for reporting unsafe practices or for suggesting pollution prevention improvements. This ensures that housekeeping becomes a collective responsibility, rather than the task of a few individuals.

Practical examples illustrate the importance of housekeeping. In tanneries, installing drip trays under drums and segregating chrome-laden waste streams significantly reduces chromium contamination in wastewater. In chemical industries, regular preventive maintenance of pumps and pipelines prevents solvent losses and reduces volatile organic compound (VOC) emissions. Similarly, in the food industry, adopting First-In-First-Out (FIFO) storage practices prevents material spoilage, while routine cleaning minimizes microbial contamination. These cases highlight how simple housekeeping measures can have a major impact on pollution reduction and resource efficiency.



Waste segregation

Waste segregation is a key element of waste management that contributes to environmental protection, natural resource conservation, and reducing the amount of waste sent to landfills. Proper segregation allows for more efficient recycling, reducing the negative impact on our planet. In this guide, we present the most important rules and tips for waste segregation.

Types of Waste for Segregation

Waste segregation is essential in providing sustainable management of waste and protecting the environment. Correct classification of different wastes into their appropriate categories is crucial because it is used in the right management of wastes in a way that is beneficial for mankind and society as a whole. Proper waste segregation minimizes the ecological damage by making sure that waste is appropriately disposed of. The following are the main types of waste to separate:

a) Organic Waste

Organic waste includes food waste, garden waste, and many more biodegradable materials. Segregation of waste management ensures that organic waste is properly identified and composted.

b) Recyclable Waste

Recyclable materials include paper, cardboard, plastic, glass, and metals. Proper rinsing and cleaning of these materials before discarding them in recycling bins minimizes contamination.

c) Non-Recyclable Waste

Some scraps, such as used tissues, plastic products (plastic wraps), and other non-recyclables, fall under non-biodegradable waste. These materials cannot be recycled and should be disposed of properly in general waste bins to prevent environmental harm.

d) Hazardous Waste

Hazardous waste includes chemicals, fluorescent bulbs, medical wastes and used car batteries. These categories of waste items would need to be taken to specific hazardous waste or collection facilities so that they do not damage the environment.

e) E-Waste (Electronic Waste)

E-waste includes discarded phones, computers, or batteries, which contain harmful chemicals that could lead to the poisoning of the earth, hence segregation is necessary.

f) Construction & Demolition Debris

Some materials, such as wood, brick, and concrete, from construction and demolition, can be recycled or reused, thus minimizing the amount of waste and saving natural resources. To get the best method to dispose of or recycle specific materials, contact local facilities.

g) Medical Waste

Used syringes, bandages, medicines are among the items that medical waste consists of. They need to be thrown away in separate bins which are clearly labeled in accordance with health and safety policy.

h) Miscellaneous Waste

Miscellaneous wastes include textiles, rubber and materials that do not fall under any of the aforementioned standard types. Handling of these would also require specific solutions for disposal as per the type of material.

Importance of waste segregation

a) Environmental Protection:

It prevents hazardous materials from contaminating soil, water, and air, reducing environmental pollution.

b) Resource Recovery:

Proper segregation ensures that recyclable and compostable materials can be efficiently collected and reused, closing pollution loops and conserving raw materials.

c) Reduced Landfill:

By diverting waste from landfills, segregation helps conserve space and reduce the generation of greenhouse gases.

d) Improved Recycling Efficiency:

Cleaner, uncontaminated waste streams lead to higher quality recycled materials and more efficient recycling processes.

e) Health and Safety:

It minimizes public exposure to harmful substances, protecting human health.

f) Cost Savings:

Segregated waste is often easier and cheaper to process than mixed waste, which requires extensive sorting.

Segregation of chromium-laden waste stream

This is a critical process for effective and safe waste management, particularly within industries like leather tanning and electroplating. The primary goal is to isolate the chromium from other waste components to facilitate proper treatment, recycling, or disposal. The method of segregation depends on the waste stream's physical state (solid or liquid) and the valence state of the chromium present, which is typically either the highly toxic hexavalent chromium Cr^{6+} or the less toxic trivalent chromium Cr^{3+} .

Segregation methods for liquid waste streams for wastewater, the segregation strategy often involves treating concentrated "chrome liquors" separately from another wastewater. This approach is more efficient and can lead to the recovery and reuse of chromium and water.

Chemical Reduction and Precipitation: A common two-stage process for treating wastewater containing toxic hexavalent chromium Cr^{6+} .

Reduction: In an acidic environment (pH 2-3), a reducing agent like sodium bisulfite or ferrous sulfate is added to convert hexavalent chromium Cr^{6+} to trivalent chromium Cr^{3+} . An ORP (oxidation-reduction potential) sensor can be used to monitor the completion of this step.

Precipitation: The pH is then raised to an alkaline level (pH 8-10) by adding a base like lime (calcium hydroxide) or caustic soda. This causes the less soluble trivalent chromium Cr^{3+} to precipitate as a solid—chromium hydroxide $\text{Cr}(\text{OH})_3$ which can then be filtered out.

Centralized Chrome Recovery: Common Effluent Treatment Plants (CETPs) may implement dedicated systems to recover chromium. Segregated chrome streams are funneled to a centralized facility where the chromium is recovered as a cake for regeneration and reuse.

Membrane Filtration: Techniques like reverse osmosis (RO) and nanofiltration (NF) use semi-permeable membranes to separate dissolved chromium ions. These systems can produce a chromium-rich concentrate (retentate) that can be reused in the process.

- I. **Ion Exchange:** This process uses specialized resin beads to capture chromium ions from the waste stream. It is effective for recovering and reusing chromium from low-concentration waste streams, but the resins must be regularly regenerated.
- II. **Bioelectrochemical Systems:** A more advanced method that uses biological processes to convert toxic hexavalent chromium to the non-toxic trivalent form while also producing electricity. Segregation methods for solid waste streams Solid waste containing chromium, such as leather shavings or spent catalysts, must be managed to prevent leaching into the environment.
- III. **Waste Characterization:** The waste is first analyzed to determine the type of chromium and other contaminants present. This is crucial for selecting the appropriate treatment and disposal method. **Physical Separation:** For mixed solid waste, manual or automated separation can remove non-hazardous materials from chromium-containing ones. For example, in the leather industry, shavings containing chromium are segregated from other organic waste.
- IV. **Hydrothermal Separation:** Innovative methods have been developed to separate and recover chromium from chromium-containing solid residues. These involve adding a mineralizer and using a hydrothermal or heat treatment process to separate the chromium from other components like aluminum.
- V. **Incineration with Control:** While incineration can remove organic content, it can also oxidize trivalent chromium to the more harmful hexavalent state if not done properly. This method requires sophisticated equipment for complete oxidation and careful handling of the resulting ash to prevent further contamination.

Reduction and reuse of chemicals in tannery effluent

Numerous research papers and technical reports detail methods for reducing and reusing chemicals in tannery effluent. The most effective strategies involve a combination of cleaner production techniques within the tannery and advanced effluent treatment technologies.

I. In-house cleaner production techniques

Cleaner production focuses on minimizing waste and chemical use at the source before the effluent is even created.

- **Advanced chrome management:** High exhaustion chrome tanning can increase chrome uptake by the leather to as much as 98%, significantly lowering the amount of chrome in the waste floats.

- Hair-save liming: Instead of dissolving the hair, this process separates it in an intact state, which reduces the chemical and organic load in the wastewater.
- Low-ammonia or ammonia-free deliming and bating: Using alternative agents instead of ammonium salts helps reduce the total nitrogen load in the effluent.
- Enzymatic treatments: Enzymes can be used for processes like unhairing and bating, reducing the need for harsh chemicals.
- Segregation of wastewater streams: Keeping different wastewater streams separate (e.g., chrome and sulfide-bearing effluents) allows for targeted, more efficient treatment and recycling.
- Reuse of spent liquors: Treating and reusing spent pickling and chrome tanning liquors can reduce the need for fresh chemicals. For example, adding sulfuric and formic acid to spent pickling liquor and basic chromium sulfate to used chrome liquor can allow for their reuse.

II. Effluent treatment and reuse technologies

These methods treat the wastewater after production, allowing for the recovery and reuse of chemicals and water.

- Membrane filtration: Advanced membrane processes like nanofiltration (NF) and reverse osmosis (RO) can produce high-quality water for reuse by removing salts and recalcitrant organic compounds.
- Membrane bioreactors (MBR): These combine a biological treatment process with membrane filtration to achieve very high removal of organic pollutants and suspended solids, producing a quality effluent suitable for recycling.
- Electrochemical treatment: Using processes like electrocoagulation and electro-oxidation can effectively remove chromium and other pollutants. The recovered chromium can sometimes be reused as a tanning agent.
- Adsorption: Using low-cost adsorbents made from materials like coffee husks, clay, or modified tannery waste can remove chromium, dyes, and other pollutants.
- Advanced oxidation processes (AOPs): Technologies like Fenton's oxidation use highly reactive radicals to break down persistent and non-biodegradable organic pollutants, making the water more suitable for further biological treatment.
- Coagulation/flocculation and dissolved air flotation (DAF): This physicochemical treatment can remove a high percentage of suspended solids, chromium, and organic matter from the effluent.

Examples of chemical recovery

- Chromium recovery: One study demonstrated that recovering chromium from tanning wastewater via chemical precipitation or electrocoagulation (using copper electrodes)

resulted in a recovered chrome-containing liquor that could be successfully used for tanning.

- Sulfide and protein recovery: Oxidizing sulfides in the unhairing effluent can reduce the overall sulfide concentration. Proteins can then be recovered by acidifying the waste.

Many of these methods can be used in combination to achieve the most effective results for reducing and reusing chemicals in tannery effluent.

Water footprint

A water footprint refers to the total amount of freshwater consumed and polluted in the production of goods and services used by an individual, community, or organization. It considers both direct water consumption and indirect water use throughout the entire production chain. The water footprint consists of three components: blue water, which represents surface and groundwater; green water, which refers to rainwater stored in the soil; and grey water, which is the amount of water required to assimilate pollutants to an acceptable quality level. This concept helps assess and manage the impact of human activities on water resources, particularly under conditions of increasing global water scarcity.

In the leather industry, the water footprint represents the total freshwater consumed and polluted throughout the leather production supply chain, beginning with animal rearing and extending through processing at the tannery. It includes blue, green, and grey water components and therefore provides a comprehensive assessment of the industry's water use and pollution. Evaluating the water footprint can help identify opportunities for reducing water consumption, controlling pollution, and minimizing the overall environmental impact of leather production.



Green Water Footprint

The water held in the unsaturated zone, formed by precipitation available to plants
Green water :Rainwater, Including soil intercepted moisture



Blue Water Footprint

Volume of surface and ground water consumed to produce a product
Blue water: Waters in lakes, rivers and aquifers



Grey Water Footprint

The amount of water required to dilute a certain amount of pollution
Grey water: Polluted fresh water

Components of the Leather Industry's Water Footprint

The water footprint of leather is broken down into three categories:

- Blue Water Footprint:

This is the amount of surface and groundwater consumed during the entire production process, including the water used in animal husbandry and the processes at the tannery itself.

- Green Water Footprint:

This refers to the rainwater consumed, primarily during the growth of the animal's feed, and the water stored in the soil that evaporates or transpires from plants.

- Grey Water Footprint:

This is the volume of freshwater needed to dilute and assimilate pollutants (like those from tanning chemicals) in wastewater to meet established water quality standards.

According to a 2023 study from Bangladesh, the total water footprint for one ton of bovine crust leather is approximately 34,000 cubic meters. The largest proportion comes from the animal's life on the farm, with the grey water footprint being particularly significant due to pollution from both farming and processing. The wet processing stage accounts for 97% of the tannery's water footprint. The largest water-consuming and polluting steps are:

- Beamhouse operations: Soaking and liming are the highest contributors to both water consumption and pollution load. These stages produce effluent with high levels of organic matter, salt, and other pollutants.
- Post-tanning stages: The dyeing stage has a high pollution contribution. Other processes like wet-back, rechroming, and finishing also add significantly to the overall water footprint.

The water footprint of a tannery includes the water used in processing and the larger, more significant footprint from raising the animals. The tanning process uses large volumes of fresh water, with estimates ranging from 35 to 60 cubic meters (35,000–60,000 liters) of water per ton of raw hides. The majority of water is used during the wet processing stages and released as highly polluting effluent.

Table 1. Water consumption rate (%) for leather production in tanneries in Bangladesh.

Main Stages	Sub Stages	Consumption Rate (%)
Soaking	Pre soaking	300% of raw weight ^a
	Main soaking	300% of raw weight
	Washing	100% of raw weight
Liming	Bath	300% of raw weight
	Washing	600% of raw weight
Fleshing and unhairing		3 kg of per 15 sq ft ^e
Chemical Wash	Washing	200% of pelt weight ^b
Deliming and Bating	Bath	100% of pelt weight
Pickling and Tanning	Washing	300% of pelt weight
	Bath	100% of pelt weight
Wet back	Bath	200% of wet blue shaved weight ^c
Rechroming	Washing	200% of wet blue shaved weight
	Bath	100% of wet blue shaved weight
Neutralization	Washing	200% of wet blue shaved weight
	Bath	150% of wet blue shaved weight
Retanning	Washing	150% of wet blue shaved weight
	Bath	150% of wet blue shaved weight
Fatliquoring	Bath	150% of wet blue shaved weight
Dyeing	Bath	200% of wet blue shaved weight
	Top dyeing	200% of wet blue shaved weight
	Washing	200% of wet blue shaved weight
Finishing		0.02 kg of per 1 sq ft ^e crust leather ^d

Note(s): (A Standard consumption data has been followed for the purpose of simplification, helped by Bangladesh Tanners Association). ^a raw weight is the weight of raw hide and skin. ^b pelt weight is the weight of the soaked and limed hide and skin. ^c wet blue shaved weight is the weight of the processed hide and skin through beam house and tan yard operations. ^d crust leather is wet blue leather processed by post tanning processes. ^e sq ft means square feet.

Environmental impacts of the water footprint

The large water footprint of tanneries has several major environmental consequences, especially if untreated effluent is discharged directly into water bodies:

- Contamination of water sources: Effluent contains high concentrations of hazardous chemicals, including chromium, sulfides, salts, and heavy metals. This can severely pollute local rivers, streams, and groundwater.
- Threat to aquatic life: The high pollutant load, which includes low oxygen levels (high BOD/COD), can be toxic to aquatic ecosystems.
- Health risks: Communities near tanneries, particularly in developing countries, may suffer from diseases due to exposure to contaminated water.